Guna Bala D

+91 7904692655 — gunad202004@gmail.com — linkedin.com/in/guna-bala-a00457374 — github.com/GunaBala-D

Education

Rajiv Gandhi College of Engineering and Technology 2021 – 2025
B.Tech in Information Technology (Graduated with A Grade) *Puducherry, India*

Experience

Srikesh Infotech (SKIT) Sep 2025 – Present
AI/ML Engineer *Puducherry, India*

- **Built an internal Accounting Co-Pilot** using Anthropic Claude and self-hosted Ollama as interchangeable LLM backends for natural-language queries over live business records; adopted across **7 internal teams**.
- **Designed a Retrieval-Augmented Generation (RAG) pipeline** on PostgreSQL/pgvector to ground LLM responses in **1,000+** accounting and operational records, using document chunking and re-ranking to improve retrieval precision.
- **Built an OCR pipeline** (pdfplumber, pytesseract) to digitize source documents for ingestion into the retrieval system.
- **Developed a LangGraph agentic workflow** that retrieves similar past incidents from ChromaDB to ground anomaly explanations, integrating **Model Context Protocol (MCP)** to connect the agent to external tools and data sources.
- **Built an unsupervised anomaly-detection pipeline** (Isolation Forest, Local Outlier Factor) over **15,000+** daily log events; tuned with grid and random search to ∼**90% precision,** ∼**88% recall,** ∼**89% F1**, surfacing results in a live dashboard for real-time monitoring.
- **Re-architected a script-based ML tool into a production FastAPI microservice** with modular inference endpoints on AWS EC2, reducing manual triage time by an estimated **40–50%**.
- Maintained production **Odoo ERP** modules (Python, Odoo ORM, OWL) alongside AI/ML development.

Twilight IT Solution Jun 2025
Web Development Intern

- Built a **customer-facing website** using **HTML, CSS, and JavaScript**, translating design requirements into a responsive, cross-browser front end.
- Implemented interactive UI components and page layouts with vanilla JavaScript, gaining hands-on experience in front-end development fundamentals.

Projects

Enterprise RAG Knowledge Assistant | *Python, FastAPI, ChromaDB, PostgreSQL, Ollama, Gemini*

- Built a **RAG knowledge assistant** with semantic chunking and Top-K retrieval over ChromaDB, returning source citations with every answer to reduce hallucination risk.
- Implemented a **FastAPI** backend with streaming responses, retrieval visibility, and user-feedback capture; persisted conversation history in **PostgreSQL** for auditability.

Real-Time Person Detection | *Python, YOLO, OpenCV*

- Built a **real-time computer vision pipeline** with YOLO and OpenCV on continuous camera feeds, applying confidence thresholding to reduce false-positive detections.

Certifications

- **Introduction to Large Language Models** – Google
- **AWS for Beginners** – AWS
- **Apply AI at Work Pathway** – OpenAI Academy (Sep 2026)

Technical Skills

Languages: Python, SQL
Generative AI: LLM Integration (Anthropic Claude, Ollama, Gemini), Prompt Engineering, RAG, Embeddings, Semantic Search, Chunking, Re-ranking, Hybrid Search, LangGraph, Model Context Protocol (MCP), Agentic Workflows
Machine Learning: scikit-learn, Pandas, NumPy, Supervised & Unsupervised Learning (Isolation Forest, Local Outlier Factor), Feature Engineering, Model Evaluation, Hyperparameter Tuning
Deep Learning & Computer Vision: PyTorch, YOLO, OpenCV
Backend & Databases: FastAPI, REST APIs, Microservices, Async Programming, PostgreSQL, pgvector, ChromaDB
Cloud & Tools: AWS (EC2), Docker, Git/GitHub, OCR (pdfplumber, pytesseract)
Languages Spoken: Tamil (Native), English (Professional Proficiency)